

Reproducing the Properties of Winning Iterated Prisoner’s Dilemma Strategies: Validation, Limitations, and Insights

João Mendes

102079

Instituto Superior Técnico
Portugal

Sara Rijo

102507

Instituto Superior Técnico
Portugal

Sofia Paiva

102835

Instituto Superior Técnico
Portugal

1 Abstract

This study reproduces and extends the analysis of winning Iterated Prisoner’s Dilemma (IPD) strategies by Glynatsi et al. [1], validating key behavioral traits while testing robustness under dynamic payoff structures and environmental uncertainty. Using the Axelrod-Python library, we simulated 20 strategically diverse agents across four tournament types (standard, noisy, probabilistic ending, and probabilistic-noisy) with variable ($T > R > P > S$, $2R > T + S$) payoffs. Our results confirm that strategic retaliation (defection after opponent cooperation, DC rate) universally enhances performance ($\beta = -1.178$ to -2.150 , $p < 0.001$), while excessive forgiveness (cooperation after mutual defection, CD rate) consistently degrades rankings ($\beta = 1.128$ to 1.940 , $p < 0.001$). Crucially, environmental volatility fundamentally reshapes success: (1) Noise amplifies DC rate’s impact by 133% and favors stochasticity (β shifts from -0.020 to $+0.039$); (2) Probabilistic endings invert strategic priorities (e.g., game-length awareness utility reverses); (3) Defector rises from 8th to 1st rank under probabilistic-noisy conditions

Notably, algorithmic complexity (memory depth, sophistication) shows negligible predictive power ($|\rho| \leq 0.33$), and payoff values (R, S, T, P) exhibit minimal correlation with outcomes—demonstrating that behavioral responses outweigh incentive calibration or computational sophistication. Meta-strategies (e.g., Meta Winner) and evolved ML strategies (HMM 5, FSM 16) prove most robust, never ranking below 6th. This work establishes that core behavioral traits drive IPD success across incentive structures, but environmental adaptability is paramount for real-world multi-agent systems.

2 Introduction

The Iterated Prisoner’s Dilemma (IPD) serves as a foundational model for studying cooperation and competition in multi-agent systems. The IPD is an extension of the classic Prisoner’s Dilemma game, repeated over multiple rounds between the same agents. In each round, two agents simultaneously choose to either Cooperate (C) or Defect (D), receiving payoffs based on both choices according to the canonical payoff matrix: Temptation (T) for defecting when the other cooperates (unilateral defection), Reward (R) for mutual cooperation, Punishment (P) for mutual defection, and Sucker’s payoff (S) for cooperating while the other defects. The ‘iterated’ nature, where agents face each other repeatedly, introduces the possibility of strategies that use the history of interactions to encourage cooperation or exploit opponents through reciprocity, punishment, and forgiveness.

Historically, Axelrod’s seminal tournaments in the 1980s identified simple strategies like Tit-for-Tat (TFT) – which cooperates first and then copies the opponent’s previous move – as robust performers. However, recent advances have dramatically expanded

the known strategy space. Despite this progress, the question of what universally defines a “winning” IPD strategy across diverse environments remains incompletely answered. The article “Properties of Winning Iterated Prisoner’s Dilemma Strategies” [1] offers a comprehensive analysis of 195 strategies across diverse tournament settings, identifying key behavioural traits that contribute to success. Their work identified key behavioural traits—such as *be a little envious* and that *it is ok to be clever*, between others—as hallmarks of high-performance strategies, challenging previously held assumptions about which traits drive success. However, their study operated under a critical assumption: a fixed canonical payoff structure ($T = 5, R = 3, P = 1, S = 0$). This assumption limits the study’s applicability to real-world multi-agent systems (MAS), where payoffs are often context-dependent. Crucially, while ZD (Zero-Determinant) strategies could be included, their mathematical reliance on specific payoff ratios raises unanswered questions about robustness under varied incentive structures. Reproducing this study is motivated by the need to validate its findings, ensure reproducibility, and explore the robustness of its conclusions under varying conditions. Previous research has often been limited by specific tournament configurations, underscoring the relevance of this extensive evaluation.

3 Approach

The reproduction will employ the Axelrod-Python library, an open-source framework designed for simulating IPD tournaments. This choice ensures consistency with the original study and leverages a well-supported platform for extensibility.

The visualization of the results is done using the matplotlib and seaborn library, which are some of the most popular for creating many kinds of visualization in Python.

Each strategy is modeled as an autonomous agent capable of making decisions based on historical interactions. The system simulates tournaments where these agents engage in repeated IPD games, allowing for the observation of emergent behaviours and performance metrics.

3.1 Strategy Selection

As previously mentioned, our simulation will take into account strategies previously coded on the Axelrod-Python library, however, computational constraints forced us to stay focused on only a subset of strategies for this reproduction.

While we acknowledge this reduction introduces potential sampling bias, this approach aligns with pedagogical recommendations while maintaining scientific rigour through choosing a subset of 20 strategies that was determined through stratified sampling across multiple behavioural classes identified by previous works.

As noted by the original paper, previous studies suffered from inadequate strategy diversity. Our approach balances computational feasibility with representative sampling across the cooperation-defection behavioural spectrum, as well as across the different memory type strategies.

3.2 Simulation Parameters

The original paper supports several tournament types, and this work considers, as well, standard, noisy, probabilistic ending, and noisy probabilistic ending tournaments.

In Standard Tournaments, N strategies are chosen, and each strategy plays an iterated game with n turns against all other strategies, except themselves. In Noisy Tournaments, N strategies and n turns are involved as well, but with the difference that, in each turn, there is a probability $p_n \leq 0.1$ that a player's action is flipped.

In Probabilistic Ending Tournaments, the tournament ends after a game with probability $p_e \leq 0.1$. Finally, the Noisy Probabilistic Ending Tournaments include both the p_n probability that a player's action is flipped and the p_e probability that the tournament ends.

Parameters p_e and p_n followed Glynatsi et al.'s [1] design: noise probability $p_n \leq 0.1$ and ending probability $p_e \leq 0.1$ to ensure interpretability.

Each tournament is repeated k times, defined randomly, and a fixed seed is used in order for the results to be reproducible. The winner of each tournament is determined based on the median score achieved by a strategy from the entire set of repetitions, not by the number of wins, since analysing high-score strategies that did not win is also interesting for the work. We also take into consideration a normalized rank, between 0 and 1 (0 = best), for each strategy of a tournament.

3.3 Payoff Structure

As identified in the proposal, the original study makes the assumption of a static payoff matrix: $T = 5, R = 3, P = 1, S = 0$

This aligns with canonical IPD literature, in satisfying $T > R > P > S$ and $2R > T + S$ as well as the specific restraint of the ZD (Zero Determinant) strategies that is that $T + S \neq R + P$, usually used as $T + S > R + P$, due to problems that may arise due to the properties of ZD strategies.

The fixed payoff matrix assumption, however, introduces a key constraint, since conclusions may not hold under alternative or dynamic ratios, that, nevertheless, cover the restraints above described.

To address these limitations, our replication tested an alternative version, with dynamic payoff values, that nevertheless, follows the constraints previously mentioned by the IPD.

However, we found that the constraints imposed by ZD strategies overly restrict how different payoff values influence outcomes. To explore more diverse results, we excluded ZD strategies from our simulations and instead evaluated strategies from all other classes without these limitations.

4 Performance Metrics and Validation Protocol

We used several performance metrics and validation protocols, most being the same or similar to the ones used in the work of Glynatsi et al. [1] such as (1) Strategy Performance, that exposes the median

normalised rank across different opponents and tournament types; (2) Spearman Correlations, that calculate the correlation between normalised rank/median score of each strategy with their features (properties of the strategies and the variables of the tournaments) and (3) Linear Regression, where we try to find features that demonstrate statistically significant predictive capability over normalised rank.

5 Results

Table 1: Strategies Ordered by Median Normalized Rank in Each Tournament Type (Rank from 0 to 1)

Standard	Noisy
DoubleCrossover (0.000)	Evolved HMM 5 (0.062)
Evolved HMM 5 (0.067)	Grudger (0.067)
Grudger (0.083)	DoubleCrossover (0.083)
Evolved FSM 16 (0.133)	Defector (0.143)
Meta Winner (0.286)	Meta Winner (0.211)
Tit For Tat (0.294)	Evolved FSM 16 (0.222)
Soft Joss (0.333)	Tit For Tat (0.333)
Defector (0.380)	Soft Joss (0.400)
Win-Stay Lose-Shift (0.389)	Suspicious Tit For Tat (0.417)
GTFT (0.462)	Win-Stay Lose-Shift (0.526)
Tit For 2 Tats (0.500)	GTFT (0.571)
Suspicious Tit For Tat (0.529)	Tit For 2 Tats (0.579)
Meta Mixer (0.667)	Meta Mixer (0.600)
Cycler DC (0.750)	Alternator (0.714)
Cooperator (0.750)	Cycler DC (0.722)
Alternator (0.750)	Random (0.722)
Random (0.778)	Win-Shift Lose-Stay (0.833)
Bully (0.882)	Bully (0.889)
Win-Shift Lose-Stay (0.923)	Anti Tit For Tat (0.947)
Anti Tit For Tat (1.000)	Cooperator (1.000)

Probabilistic	Prob Noisy
Grudger (0.000)	Defector (0.000)
Evolved HMM 5 (0.105)	Grudger (0.059)
Evolved FSM 16 (0.114)	Evolved HMM 5 (0.125)
Defector (0.125)	Evolved FSM 16 (0.188)
DoubleCrossover (0.250)	Suspicious Tit For Tat (0.278)
Tit For Tat (0.333)	Meta Winner (0.286)
Meta Winner (0.375)	DoubleCrossover (0.333)
Soft Joss (0.389)	Tit For Tat (0.389)
Win-Stay Lose-Shift (0.421)	Soft Joss (0.467)
Suspicious Tit For Tat (0.429)	Win-Stay Lose-Shift (0.526)
GTFT (0.529)	Cycler DC (0.571)
Cycler DC (0.625)	Meta Mixer (0.600)
Meta Mixer (0.632)	Win-Shift Lose-Stay (0.667)
Tit For 2 Tats (0.667)	Random (0.667)
Random (0.722)	GTFT (0.725)
Alternator (0.765)	Alternator (0.727)
Bully (0.800)	Bully (0.778)
Win-Shift Lose-Stay (0.833)	Tit For 2 Tats (0.833)
Cooperator (0.944)	Anti Tit For Tat (0.941)
Anti Tit For Tat (1.000)	Cooperator (1.000)

The normalized rank analysis reveals significant shifts in strategy effectiveness across tournament conditions. Grudger and Double-Crossover dominated standard tournaments, achieving near-perfect median ranks (0.000 and 0.083 respectively). Evolved machine learning strategies (HMM 5 and FSM 16) consistently ranked among the top five performers in all environments, with HMM 5 maintaining elite status (0.062–0.125) even under noisy conditions. This stability contrasts sharply with Tit For Tat, which degraded from 6th position in standard tournaments to 8th in noisy environments.

Notably, Defector's performance exhibited extreme environment-dependence. While ranking poorly in standard tournaments (8th at 0.380), it became a top-tier strategy in probabilistic-noisy conditions (1st at 0.000). Similarly, Suspicious Tit For Tat improved from 12th to

4th place when noise was introduced. These reversals highlight how environmental uncertainty fundamentally reshapes competitive hierarchies.

Cooperator and Anti Tit For Tat consistently ranked worst across all conditions, with Anti Tit For Tat achieving the lowest possible rank (1.000) in three of four tournaments. This suggests unconditional cooperation and spiteful defection represent evolutionarily unstable strategies regardless of environmental perturbations.

The probabilistic ending mechanism introduced unique performance patterns: Win-Stay Lose-Shift maintained mid-tier rankings (9th–10th) while Cyclus DC improved from 15th in standard tournaments to 11th in probabilistic settings. Meta-strategies (Meta Winner, Meta Mixer) demonstrated remarkable robustness, never ranking below 6th place in any environment—indicating adaptive rule-sets outperform fixed behavioral programs under variable conditions.

As previously mentioned, we produced Spearman Correlations for each strategy in each tournament style. More specifically, the correlation between normalised rank/median score of each strategy with their features. Features, in this case, refers to variables of the tournaments, such as payoffs, and to the properties of the strategies, such as the memory and others.

Standard Tournament			Noisy Tournament		
	normalized_rank	Median_score		normalized_rank	Median_score
Cooperation_rating	0.072	0.034	Cooperation_rating	0.54	-0.3
Initial_C_rate	-0.39	0.37	Initial_C_rate	-0.22	0.22
CC_rate	-0.34	0.4	CC_rate	0.056	0.078
CD_rate	0.78	-0.68	CD_rate	0.85	-0.64
DC_rate	-0.24	0.19	DC_rate	-0.56	0.43
DD_rate	0.061	-0.19	DD_rate	-0.42	0.15
CC_to_C_rate	-0.44	0.41	CC_to_C_rate	-0.22	0.19
CD_to_C_rate	0.29	-0.21	CD_to_C_rate	0.42	-0.26
DC_to_C_rate	0.098	-0.099	DC_to_C_rate	0.21	-0.14
DD_to_C_rate	0.35	-0.32	DD_to_C_rate	0.56	-0.39
R	0	0.38	R	0	0.38
S	0	0.23	S	0	0.3
T	0	0.24	T	0	0.3
P	0	0.22	P	0	0.32
noise	0	0.016	noise	0	-0.08
prob_end	0	0.023	prob_end	0	0.041
repetitions	0	0.017	repetitions	0	0.019
turns	0	-0.027	turns	0	-0.1
n_strategies	0	-0.052	n_strategies	0	-0.056
coop_mean	0	0.22	coop_mean	0	0.25
coop_median	0	0.17	coop_median	0	0.23
coop_min	0	0.14	coop_min	0	0.064
coop_max	0	0.06	coop_max	0	0.08
coop_vs_mean	0.096	-0.089	coop_vs_mean	0.59	-0.46
coop_vs_median	0.093	-0.066	coop_vs_median	0.58	-0.41
min_vs_coop	-0.021	0.14	min_vs_coop	-0.31	0.27
coop_vs_max	0.09	-0.013	coop_vs_max	0.53	-0.34
memory_depth	-0.19	0.2	memory_depth	-0.16	0.2
stochastic	-0.12	0.091	stochastic	-0.12	0.09
uses_length	-0.31	0.25	uses_length	-0.33	0.25
uses_game	-0.11	0.085	uses_game	-0.17	0.13

Figure 1: Resulting Spearman Correlations, Standard and Noisy Tournament types, in randomly generated payoff values

Probabilistic Tournament			Prob. noisy Tournament		
	normalized_rank	Median_score		normalized_rank	Median_score
Cooperation_rating	0.17	-0.0071	Cooperation_rating	0.51	-0.19
Initial_C_rate	-0.12	0.15	Initial_C_rate	-0.00023	0.063
CC_rate	-0.14	0.24	CC_rate	0.11	0.071
CD_rate	0.72	-0.54	CD_rate	0.74	-0.49
DC_rate	-0.28	0.16	DC_rate	-0.55	0.33
DD_rate	-0.049	-0.13	DD_rate	-0.38	0.049
CC_to_C_rate	-0.27	0.24	CC_to_C_rate	-0.078	0.09
CD_to_C_rate	0.38	-0.24	CD_to_C_rate	0.41	-0.22
DC_to_C_rate	0.083	-0.063	DC_to_C_rate	0.21	-0.1
DD_to_C_rate	0.28	-0.21	DD_to_C_rate	0.51	-0.28
R	0	0.51	R	0	0.51
S	0	0.29	S	0	0.34
T	0	0.32	T	0	0.37
P	0	0.25	P	0	0.31
noise	0	0.018	noise	0	-0.072
prob_end	0	0.097	prob_end	0	0.16
repetitions	0	0.022	repetitions	0	0.023
turns	0	-0.011	turns	0	-0.036
n_strategies	0	-0.047	n_strategies	0	-0.046
coop_mean	0	0.26	coop_mean	0	0.29
coop_median	0	0.2	coop_median	0	0.27
coop_min	0	0.17	coop_min	0	0.11
coop_max	0	0.078	coop_max	0	0.091
coop_vs_mean	0.2	-0.14	coop_vs_mean	0.56	-0.36
coop_vs_median	0.18	-0.11	coop_vs_median	0.55	-0.32
min_vs_coop	-0.044	0.18	min_vs_coop	-0.26	0.23
coop_vs_max	0.19	-0.06	coop_vs_max	0.5	-0.25
memory_depth	-0.14	0.12	memory_depth	-0.11	0.09
stochastic	-0.059	0.047	stochastic	-0.033	0.032
uses_length	-0.13	0.096	uses_length	-0.13	0.078
uses_game	-0.06	0.025	uses_game	-0.1	0.048

Figure 2: Resulting Spearman Correlations, Probabilistic Ending and Noisy Probabilistic Ending Tournament types in randomly generated payoff values

Spearman correlation analysis revealed consistent behavioural predictors of strategy performance across tournament environments. The strongest predictor was a strategy's cooperation rate after defection (CD rate), which showed a robust positive correlation with normalized rank ($\rho \in [0.72, 0.85]$ across all tournaments). This indicates that strategies cooperating after mutual defection consistently achieved worse rankings, with the effect intensifying in noisy environments.

Conversely, defection after opponent cooperation (DC rate) exhibited significant negative correlations, although more dispersed, ($\rho \in [-0.24, -0.56]$), suggesting that 'punishing' cooperative opponents reduced performance—particularly under noisy conditions where this relationship strengthened by 133%. Notably, cooperation after mutual cooperation (CC to C rate) showed consistently negative associations ($\rho \in [-0.22, 0.44]$), implying that persistent cooperation following mutual cooperation improved rankings.

As we can observe, noise fundamentally altered predictor importance: overall cooperation rates (Cooperation rating) shifted from weak correlation in standard tournaments ($\rho = 0.07$) to strong positive correlation in noisy settings ($\rho \in [0.51, 0.54]$). Similarly, relative cooperation (coop vs median) became a key predictor in noisy environments ($\rho \in [0.55, 0.59]$). Surprisingly, strategy sophistication metrics like memory depth and stochasticity showed minimal predictive power ($|\rho| \leq 0.33$), as did game parameters

(R/S/T/P values), indicating behavioural responses outweigh algorithmic complexity in determining competitive success.

Table 2: OLS Regression Results for Normalized Rank Across Tournament Types

Variable	Standard		Noisy		Probabilistic		Prob-Noisy	
	Coeff.	p	Coeff.	p	Coeff.	p	Coeff.	p
<i>Key Behavioral Traits</i>								
CD_rate	1.128	<0.001	1.111	<0.001	1.551	<0.001	1.940	<0.001
DC_rate	-1.178	<0.001	-1.367	<0.001	-1.664	<0.001	-2.150	<0.001
DD_rate	0.677	<0.001	0.785	<0.001	0.707	<0.001	0.572	<0.001
CC_rate	-0.704	<0.001	-0.810	<0.001	-0.569	<0.001	-0.432	0.004
Cooperation_rating	0.424	<0.001	0.301	<0.001	0.423	0.016	-0.156	0.540
<i>Transition Probabilities</i>								
CD_to_C_rate	0.116	<0.001	0.131	<0.001	0.102	<0.001	0.067	<0.001
DC_to_C_rate	0.041	<0.001	0.058	<0.001	0.043	<0.001	0.040	<0.001
DD_to_C_rate	0.054	<0.001	0.124	<0.001	0.036	<0.001	0.067	<0.001
CC_to_C_rate	0.045	<0.001	-0.047	<0.001	0.032	0.003	-0.047	<0.001
<i>Cooperation Metrics</i>								
coop_mean	1.119	<0.001	1.081	<0.001	0.657	<0.001	0.665	<0.001
coop_vs_mean	-0.006	0.861	-0.016	0.613	-0.351	<0.001	-0.333	<0.001
coop_vs_median	-0.005	0.640	-0.027	0.003	0.048	0.162	-0.033	0.006
coop_max	-0.173	<0.001	0.077	0.025	0.041	0.461	0.356	<0.001
<i>Strategy Properties</i>								
memory_depth	-0.010	0.020	-0.015	0.001	-0.025	<0.001	-0.029	<0.001
stochastic	-0.020	<0.001	0.015	<0.001	0.002	0.657	0.039	<0.001
uses_length	-0.091	<0.001	-0.058	<0.001	0.084	<0.001	0.112	<0.001
uses_game	0.042	<0.001	0.075	<0.001	-0.064	<0.001	-0.076	<0.001

Regression analysis established that cooperating after defection (CD rate) consistently degraded strategy rankings across all tournament types ($\beta = 1.128$ to 1.940 , $p < 0.001$), while defecting after opponent cooperation (DC rate) universally enhanced performance ($\beta = -1.178$ to -2.150 , $p < 0.001$). Tournaments with noise amplified these effects, with DC rate’s impact intensifying in probabilistic-noisy environments. Counter-intuitively, higher overall cooperation rates (coop mean) systematically reduced performance ($\beta = 0.657$ – 1.119 , $p < 0.001$). Noise fundamentally altered strategic requirements: stochasticity shifted from detrimental ($\beta = -0.020$, Standard) to beneficial ($\beta = +0.039$, Prob-Noisy; $p < 0.001$), while probabilistic endings reversed the utility of game-length awareness (uses length: $\beta = -0.091 \rightarrow +0.084$). Algorithmic properties (such as memory depth) showed negligible practical impact.

6 Discussion

This reproduction study validates and extends the core findings of Glynatsi et al. [1], while revealing critical nuances in how tournament environments shape the success of IPD strategies. Our results robustly confirm that behavioural traits, rather than algorithmic sophistication or rigid payoff structures, universally drive competitive performance. Specifically, strategies exhibiting strategic retaliation (defection after opponent cooperation, DC rate) consistently achieved higher rankings across all tournament types, with regression coefficients (β) ranging from -1.178 to -2.150 ($p < 0.001$). Conversely, excessive forgiveness (cooperation after mutual defection, CD rate) systematically degraded performance ($\beta = 1.128$ – 1.940 , $p < 0.001$). These effects intensified significantly under environmental uncertainty: in probabilistic-noisy tournaments, the performance impact of DC rate increased by 133% compared to standard conditions. This underscores the original study’s assertion that winning strategies must balance reciprocity with judicious punishment.

Environmental volatility fundamentally reshaped strategic hierarchies. Noise amplified the importance of adaptive behaviours: stochasticity shifted from detrimental ($\beta = -0.020$ in standard tournaments) to beneficial ($\beta = +0.039$ in probabilistic-noisy; $p < 0.001$), while overall cooperation rates became strongly correlated with poor outcomes ($\rho \in [0.51, 0.54]$). Probabilistic endings inverted strategic priorities, as evidenced by the reversal in utility for game-length awareness (uses length: $\beta = -0.091 \rightarrow +0.084$). These dynamics explain dramatic rank reversals, such as Defector’s rise from 8th place in standard tournaments to 1st under probabilistic-noisy conditions. Such sensitivity highlights the limitations of prescribing context-agnostic “winning” traits.

Critically, our exploration of dynamic payoff matrices (while preserving $T > R > P > S$ and $2R > T + S$) challenged two key assumptions. First, algorithmic complexity—including memory depth and stochasticity—showed negligible predictive power ($|\rho| \leq 0.33$), contradicting the notion that sophisticated strategies inherently dominate. Second, payoff values (R, S, T, P) themselves exhibited minimal correlation with performance (Figs. 1–2), indicating that behavioural responses outweigh incentive calibration in determining success. This suggests that the original study’s fixed-payoff conclusions hold under variable incentives, provided core IPD constraints are met.

The exceptional robustness of meta-strategies (Meta Winner, Meta Mixer) and evolved machine learning strategies (HMM 5, FSM 16)—which never ranked below 6th place—validates the original insight that adaptive rule sets outperform fixed behavioural programs. By contrast, rigid strategies like Cooperator and Anti Tit For Tat consistently failed (median rank = 1.000), confirming the evolutionary instability of unconditional cooperation or spiteful defection.

7 Limitations and Future Work

This study faces three primary limitations. First, computational constraints restricted analysis to 20 strategies, potentially underrepresenting niche behavioural classes. Second, the exclusion of Zero-Determinant (ZD) strategies—necessary to accommodate dynamic payoffs—precludes conclusions about their robustness under variable incentives. Third, tested payoff matrices adhered to canonical IPD inequalities; real-world scenarios may violate constraints like $T + S > R + P$ or assume linear payoffs. Future work should: (1) expand strategy diversity, including ZD variants under adjusted mathematical constraints; (2) model non-linear or asymmetric payoffs reflecting economic/social systems; and (3) design strategies that dynamically modulate traits like CD/DC rates based on environmental detection.

References

- [1] Nikoleta E. Glynatsi, Vincent Knight, and Marc Harper. 2024. Properties of winning Iterated Prisoner’s Dilemma strategies. *PLOS Computational Biology* 20, 12 (12 2024), 1–19. doi:10.1371/journal.pcbi.1012644